

PROGRAM

```
import javax.swing.*;
import java.awt.*;
import java.awt.event.*;

class TrafficLight extends JPanel implements ActionListener {

    private JRadioButton r1;
    private JRadioButton r2;
    private JRadioButton r3;
    private Color stop_c;
    private Color go_c;
    private Color ready_c;

    public TrafficLight() {
        setBounds(0, 0, 600, 480);

        r1 = new JRadioButton("Stop");
        r2 = new JRadioButton("Go");
        r3 = new JRadioButton("Ready");

        ButtonGroup group = new ButtonGroup(); // javax.swing package

        r1.setSelected(false);

        group.add(r1); // Add radio button to group
        group.add(r2);
        group.add(r3); // Only one radio button selected at a time

        add(r1);
        add(r2);
        add(r3); // Added to JPanel

        // Initialize colors to background color
        stop_c = getBackground();
        go_c = getBackground();
        ready_c = getBackground();

        r1.addActionListener(this); // This class implements ActionListener
```

```

    r2.addActionListener(this);
    r3.addActionListener(this);
}

public void actionPerformed(ActionEvent e) {
    if (r1.isSelected()) {
        stop_c = Color.red;
        go_c = getBackground();
        ready_c = getBackground();
    } else if (r2.isSelected()) {
        stop_c = getBackground();
        go_c = Color.green;
        ready_c = getBackground();
    } else if (r3.isSelected()) {
        stop_c = getBackground();
        go_c = getBackground();
        ready_c = Color.orange;
    }
    repaint();
}

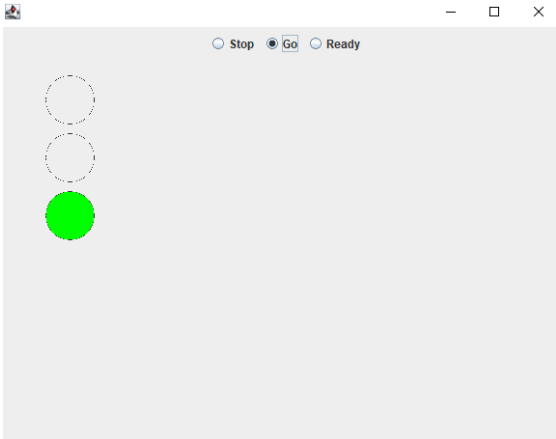
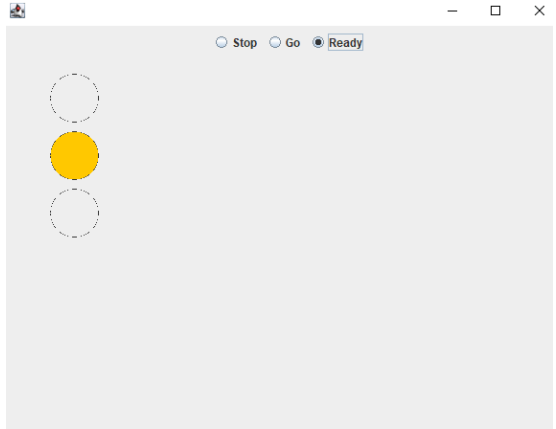
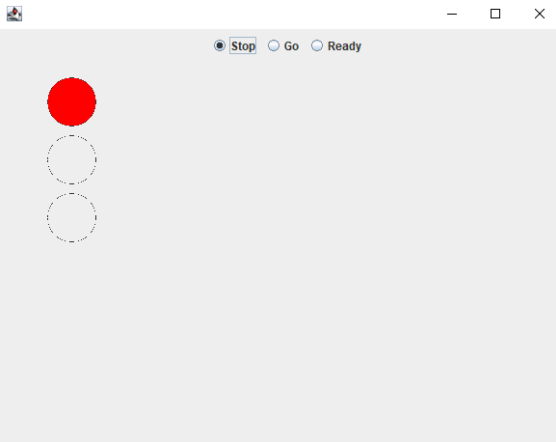
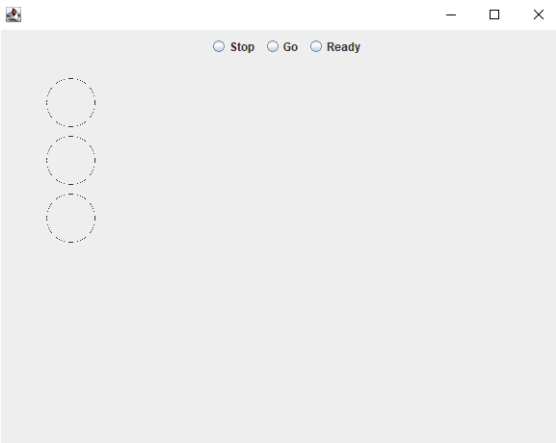
public void paintComponent(Graphics g) { // Called by repaint()
    super.paintComponent(g); // Do this before going for custom painting
    // Draw the traffic light circles
    g.drawOval(50, 50, 50, 50);
    g.drawOval(50, 110, 50, 50);
    g.drawOval(50, 170, 50, 50);

    // Fill the ovals based on the current color states
    g.setColor(stop_c);
    g.fillOval(50, 50, 50, 50);
    g.setColor(ready_c);
    g.fillOval(50, 110, 50, 50);

```

```
g.setColor(go_c);
g.fillOval(50, 170, 50, 50); // At a time one color
}
}
class TestTraffic {
    public static void main(String args[]) {
        JFrame f1 = new JFrame();
        f1.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE); // Close the application
        f1.setVisible(true);
        f1.setSize(600, 480);
        f1.setLayout(null);
        TrafficLight t = new TrafficLight();
        f1.add(t);
    }
}
```

OUTPUT



ALGORITHM

1. Select color of the traffic light using radio buttons
2. A class TrafficLight is defined.
3. It extends JPanel and implements ActionListener
4. Define variables for colors, Red, Green, Orange
5. Three radio buttons used to select color. At a time only one radio button should be activated. So group them using Button Group of javax.swing package
6. Radio button corresponding to red color is selected for initial screen.
7. When click on a radio button, for example go, make color of stop and ready traffic lights to background color of JPanel.
8. `stop_c = getBackground ();`
9. `ready_c = getBackground();`
10. `go_c = Color.green;`
11. Repaint all lights by using `repaint()` method.
12. The TrafficLight object is added to a JFrame in `main()` method